

	1	2	3	4	5	6	7	8	
	DOC. NAME	PCI EXPATION BOARD			SHEET NAME	COVER		SHEET 0 OF 14	
A	PCI EXPANTION BOARD (Date:2007.04.14)							Designed By Tae-Jin Kim	A
								Date 2007.04.14	
								CKD	
B								APPD	
								Rev No.	B
								Test DOC. No.	
								APPD Date	
C								Changed	C
D									D
E									E
F								www..ywinfocomm.com	F
								TEL:031-203-1161 FAX:031-204-3096	
	1	2	3	4	5	6	7	8	

1		2		3		4		5		6		7		8	
DOC. NAME		PCI EXPATION BOARD				SHEET NAME		TITLE				SHEET 1 OF 14			
A 															

PCI EXPANTION BOARD

- SHEET #0 : PCI EXPANTION BOARD COVER
- SHEET #1 : SHEET TABLE
- SHEET #2 : FPGA CONFIGURATION
- SHEET #3 : FPGA POWER
- SHEET #4 : FPGA BANK1/2/3/4
- SHEET #5 : FPGA BANK5/6/7/8
- SHEET #6 : FPGA CLOCK & BY-PASS
- SHEET #7 : FPGA BANK9/10
- SHEET #8 : USER I/O & RESET
- SHEET #9 : 32BIT PCI-SLOT 1/2
- SHEET #10 : 32BIT PCI-SLOT 3/4
- SHEET #11 : SYSTEM POWER
- SHEET #12 : INTERFACE CONNECTOR
- SHEET #13 : PCI POWER CONVERTER
- SHEET #14 : SYSTEM ACE

Designed By
Tae-Jin Kim

Date
2007.03.26

CKD

APPD

Rev No.

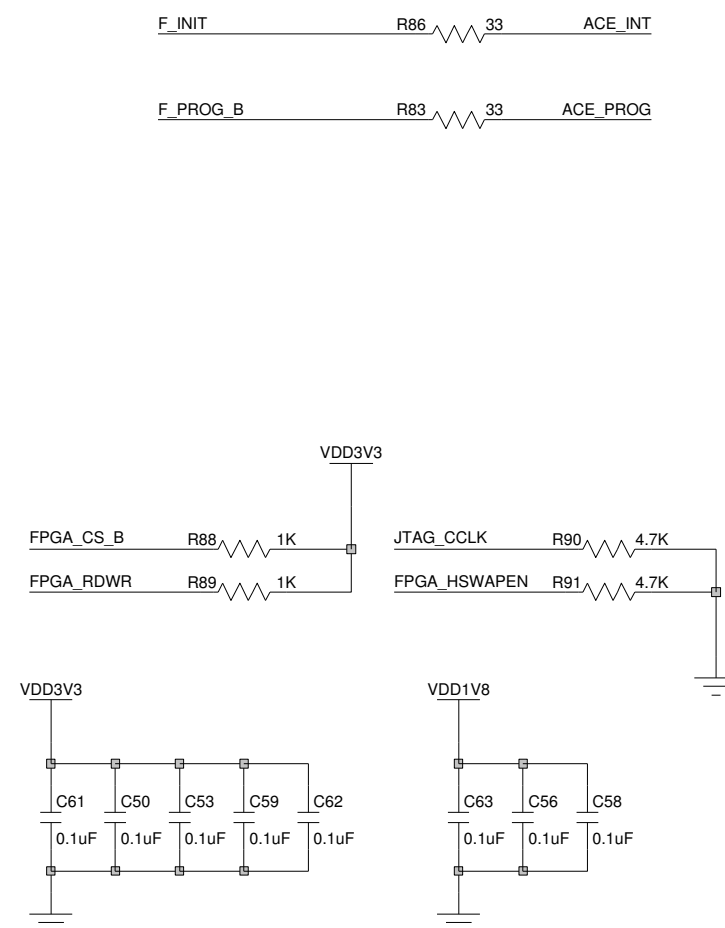
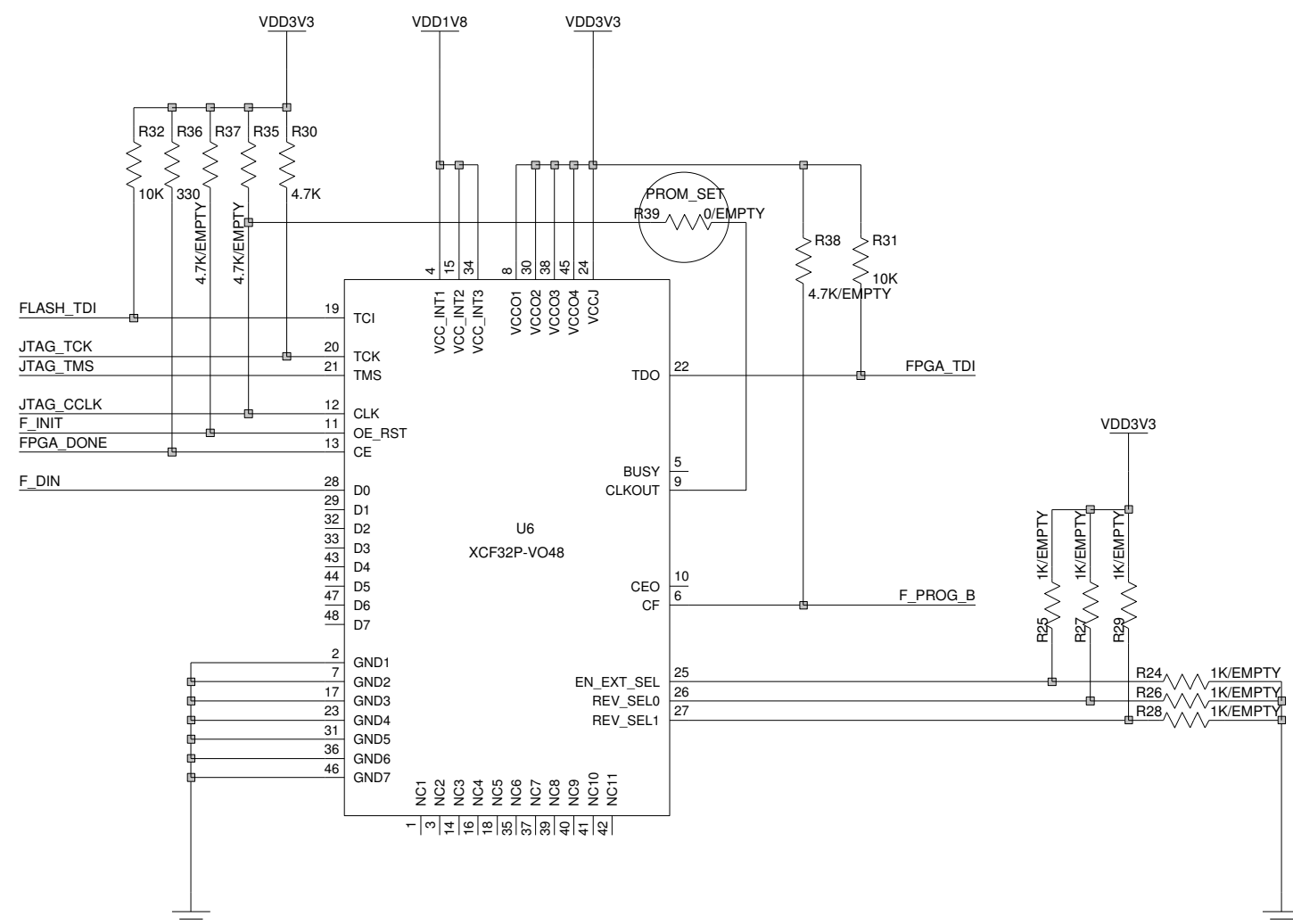
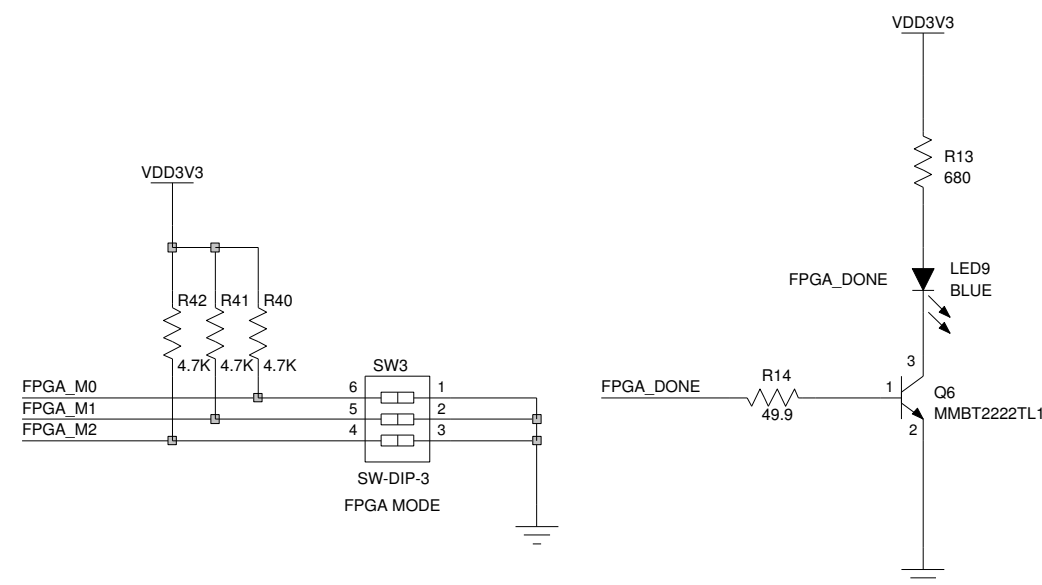
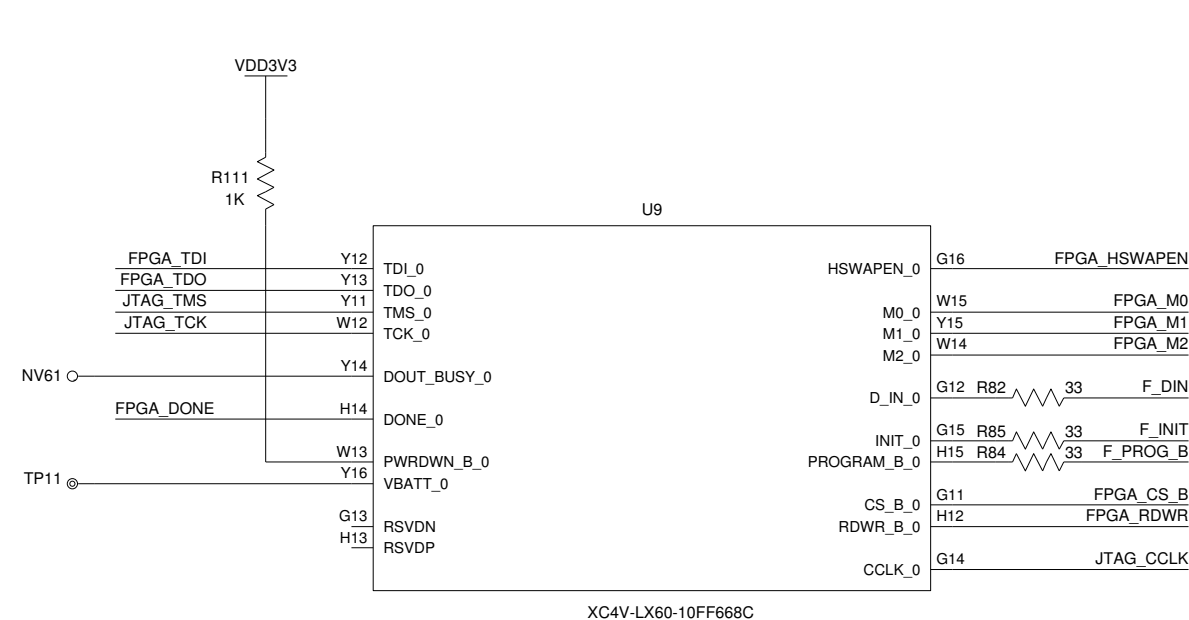
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DOC. NAME

PCI EXPANTION BOARD

SHEET NAME

FPGA Configuration

SHEET 2 OF 14

Designed By
Tae-Jin Kim

Date
2007.03.26

CKD

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Rev No.

Test DOC. No.

APPD Date

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1		2		3		4		5		6		7		8		
DOC. NAME		PCI EXPANTION BOARD						SHEET NAME		FPGA BANK 5/6/7/8				SHEET 5 OF 14		
A		Designed By Tae-Jin Kim														A
		Date 2007.03.26														
		CKD														
		APPD														
B		Rev No.														B
		Test DOC. No.														
		APPD Date														
		Changed														
C																C
D																D
E																E
F																F
1		2		3		4		5		6		7		8		

EXP_IO_(128) D10		IO_L1P_6	IO_L1P_5	C17	EXP_IO_(63) W2	IO_L1P_8	V21	TCK(0)
EXP_IO_(129) C10		IO_L1N_6	IO_L1N_5	D17	EXP_IO_(62) W1	IO_L1N_8	V22	TCK(1)
EXP_IO_(130) D9		IO_L2P_6	IO_L2P_5	C20	EXP_IO_(61) V6	IO_L2P_8	W25	TCK(2)
EXP_IO_(131) C8		IO_L2N_6	IO_L2N_5	B20	EXP_IO_(60) V5	IO_L2N_8	W26	TCK(3)
EXP_IO_(132) A8		IO_L3P_6	IO_L3P_5	B18	EXP_IO_(59) W7	IO_L3P_8	W21	TRST(0)
EXP_IO_(133) A7		IO_L3N_6	IO_L3N_5	A18	EXP_IO_(58) V7	IO_L3N_8	W22	TRST(1)
EXP_IO_(134) D8		IO_L4P_6	IO_L4P_5	D20	EXP_IO_(57) W4	IO_L4P_8	W23	TRST(2)
EXP_IO_(135) D7		IO_L4N_VREF_6	IO_L4N_VREF_5	D19	EXP_IO_(56) W3	IO_L4N_VREF_8	W24	TRST(3)
EXP_IO_(136) F10		IO_L5P_6	IO_L5P_5	E17	EXP_IO_(55) W6	IO_L5P_8	W20	TMS(0)
EXP_IO_(137) E10		IO_L5N_6	IO_L5N_5	F17	EXP_IO_(54) W5	IO_L5N_8	V20	TMS(1)
EXP_IO_(138) A6		IO_L6P_6	IO_L6P_5	C21	EXP_IO_(53) Y2	IO_L6P_8	Y25	TMS(2)
EXP_IO_(139) A5		IO_L6N_6	IO_L6N_5	B21	EXP_IO_(52) Y1	IO_L6N_8	Y26	TMS(3)
EXP_IO_(140) E9		IO_L7P_6	IO_L7P_5	C19	EXP_IO_(51) AA4	IO_L7P_8	AB24	TDI(0)
EXP_IO_(141) F9		IO_L7N_6	IO_L7N_5	D18	EXP_IO_(50) AA3	IO_L7N_8	AB25	TDI(1)
EXP_IO_(142) B6		IO_L8P_CC_LC_6	IO_L8P_CC_LC_5	A24	EXP_IO_(49) Y4	IO_L8P_CC_LC_8	AA24	TDI(2)
EXP_IO_(143) C6		IO_L8N_CC_LC_6	IO_L8N_CC_LC_5	A23	EXP_IO_(48) Y3	IO_L8N_CC_LC_8	Y24	TDI(3)
EXP_IO_(144) G10		IO_L9P_CC_LC_6	IO_L9P_CC_LC_5	G18	EXP_IO_(47) Y6	IO_L9P_CC_LC_8	AC25	
EXP_IO_(145) G9		IO_L9N_CC_LC_6	IO_L9N_CC_LC_5	G17	EXP_IO_(46) Y5	IO_L9N_CC_LC_8	AC26	
EXP_IO_(146) F8		IO_L10P_6	IO_L10P_5	B24	EXP_IO_(45) AB1	IO_L10P_8	AB26	○ NV63
EXP_IO_(147) G8		IO_L10N_6	IO_L10N_5	B23	EXP_IO_(44) AA1	IO_L10N_8	AA26	○ NV62
EXP_IO_(148) B7		IO_L11P_6	IO_L11P_5	F18	EXP_IO_(43) AC4	IO_L11P_8	AD25	○ NV68
EXP_IO_(149) C7		IO_L11N_6	IO_L11N_5	E18	EXP_IO_(42) AB4	IO_L11N_8	AD26	○ NV69
EXP_IO_(150) C5		IO_L12P_6	IO_L12P_5	E21	EXP_IO_(41) AB3	IO_L12P_8	Y22	
EXP_IO_(151) D5		IO_L12N_VREF_6	IO_L12N_VREF_5	D21	EXP_IO_(40) AB2	IO_L12N_VREF_8	Y23	
EXP_IO_(152) A9		IO_L13P_6	IO_L13P_5	A20	EXP_IO_(39) AC5	IO_L13P_8	AC22	
EXP_IO_(153) B9		IO_L13N_6	IO_L13N_5	A19	EXP_IO_(38) AB5	IO_L13N_8	AB22	
EXP_IO_(154) A3		IO_L14P_6	IO_L14P_5	D22	EXP_IO_(37) AC2	IO_L14P_8	AB23	
EXP_IO_(155) B3		IO_L14N_6	IO_L14N_5	C22	EXP_IO_(36) AC1	IO_L14N_8	AA23	
EXP_IO_(156) A4		IO_L15P_6	IO_L15P_5	A22	EXP_IO_(35) AF3	IO_L15P_8	AD22	
EXP_IO_(157) B4		IO_L15N_6	IO_L15N_5	A21	EXP_IO_(34) AE3	IO_L15N_8	AD23	
EXP_IO_(158) C4		IO_L16P_6	IO_L16P_5	D24	EXP_IO_(33) AD2	IO_L16P_8	AC23	
EXP_IO_(159) D4		IO_L16N_6	IO_L16N_5	C24	EXP_IO_(32) AD1	IO_L16N_8	AC24	
EXP_IO_(160) E7		IO_L17P_6	IO_L17P_5	G19	EXP_IO_(31) AF4	IO_L17P_8	AF19	PCI_ADDR(0)
EXP_IO_(161) D6		IO_L17N_6	IO_L17N_5	F19	EXP_IO_(30) AE4	IO_L17N_8	AF20	PCI_ADDR(1)
EXP_IO_(162) E6		IO_L18P_6	IO_L18P_5	E23	EXP_IO_(29) AD3	IO_L18P_8	Y19	PCI_ADDR(2)
EXP_IO_(163) E5		IO_L18N_6	IO_L18N_5	E22	EXP_IO_(28) AC3	IO_L18N_8	W19	PCI_ADDR(3)
EXP_IO_(164) F7		IO_L19P_6	IO_L19P_5	F20	EXP_IO_(27) AF6	IO_L19P_8	AF23	PCI_ADDR(4)
EXP_IO_(165) G7		IO_L19N_6	IO_L19N_5	E20	EXP_IO_(26) AF5	IO_L19N_8	AE23	PCI_ADDR(5)
EXP_IO_(166) C2		IO_L20P_6	IO_L20P_5	C26	EXP_IO_(25) AA7	IO_L20P_8	Y20	PCI_ADDR(6)
EXP_IO_(167) C1		IO_L20N_VREF_6	IO_L20N_VREF_5	C25	EXP_IO_(24) Y7	IO_L20N_VREF_8	Y21	PCI_ADDR(7)
EXP_IO_(168) H8		IO_L21P_6	IO_L21P_5	D23	EXP_IO_(23) AA9	IO_L21P_8	AA18	PCI_ADDR(8)
EXP_IO_(169) H7		IO_L21N_6	IO_L21N_5	C23	EXP_IO_(22) Y9	IO_L21N_8	Y18	PCI_ADDR(9)
EXP_IO_(170) D3		IO_L22P_6	IO_L22P_5	H20	EXP_IO_(21) AD5	IO_L22P_8	AF24	PCI_ADDR(10)
EXP_IO_(171) E4		IO_L22N_6	IO_L22N_5	G20	EXP_IO_(20) AD4	IO_L22N_8	AE24	PCI_ADDR(11)
EXP_IO_(172) G6		IO_L23P_VRN_6	IO_L23P_VRN_5	G22	EXP_IO_(19) AE7	IO_L23P_VRN_8	AE20	PCI_ADDR(12)
EXP_IO_(173) G5		IO_L23N_VRP_6	IO_L23N_VRP_5	G21	EXP_IO_(18) AD7	IO_L23N_VRP_8	AD20	PCI_ADDR(13)
		IO_L24P_CC_LC_6	IO_L24P_CC_LC_5	F24	EXP_IO_(17) AC6	IO_L24P_CC_LC_8	AC21	PCI_ADDR(14)
		IO_L24N_CC_LC_6	IO_L24N_CC_LC_5	F23	EXP_IO_(16) AB6	IO_L24N_CC_LC_8	AB21	PCI_ADDR(15)
		IO_L25P_CC_LC_6	IO_L25P_CC_LC_5	D26	EXP_IO_(15) AF8	IO_L25P_CC_LC_8	AD19	PCI_ADDR(16)
		IO_L25N_CC_LC_6	IO_L25N_CC_LC_5	D25	EXP_IO_(14) AF7	IO_L25N_CC_LC_8	AC19	PCI_ADDR(17)
		IO_L26P_6	IO_L26P_5	H22	EXP_IO_(13) AA8	IO_L26P_8	AA19	PCI_ADDR(18)
		IO_L26N_6	IO_L26N_5	H21	EXP_IO_(12) Y8	IO_L26N_8	AA20	PCI_ADDR(19)
		IO_L27P_6	IO_L27P_5	E25	EXP_IO_(11) Y10	IO_L27P_8	Y17	PCI_ADDR(20)
		IO_L27N_6	IO_L27N_5	E24	EXP_IO_(10) AA10	IO_L27N_8	AA17	PCI_ADDR(21)
		IO_L28P_6	IO_L28P_5	G24	EXP_IO_(9) AC7	IO_L28P_8	AB20	PCI_ADDR(22)
		IO_L28N_VREF_6	IO_L28N_VREF_5	G23	EXP_IO_(8) AB7	IO_L28N_VREF_8	AC20	PCI_ADDR(23)
		IO_L29P_6	IO_L29P_5	F26	EXP_IO_(7) AC9	IO_L29P_8	AC18	PCI_ADDR(24)
		IO_L29N_6	IO_L29N_5	E26	EXP_IO_(6) AB9	IO_L29N_8	AB18	PCI_ADDR(25)
		IO_L30P_6	IO_L30P_5	H24	EXP_IO_(5) AE6	IO_L30P_8	AF21	PCI_ADDR(26)
		IO_L30N_6	IO_L30N_5	H23	EXP_IO_(4) AD6	IO_L30N_8	AF22	PCI_ADDR(27)
		IO_L31P_6	IO_L31P_5	G26	EXP_IO_(3) AF9	IO_L31P_8	AF18	PCI_ADDR(28)
		IO_L31N_6	IO_L31N_5	G25	EXP_IO_(2) AE9	IO_L31N_8	AE18	PCI_ADDR(29)
		IO_L32P_6	IO_L32P_5	H26	EXP_IO_(1) AD8	IO_L32P_8	AE21	PCI_ADDR(30)
		IO_L32N_6	IO_L32N_5	H25	EXP_IO_(0) AC8	IO_L32N_8	AD21	PCI_ADDR(31)

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EXP_IO_(128) D10
EXP_IO_(129) C10
EXP_IO_(130) D9
EXP_IO_(131) C8
EXP_IO_(132) A8
EXP_IO_(133) A7
EXP_IO_(134) D8
EXP_IO_(135) D7
EXP_IO_(136) F10
EXP_IO_(137) E10
EXP_IO_(138) A6
EXP_IO_(139) A5
EXP_IO_(140) E9
EXP_IO_(141) F9
EXP_IO_(142) B6
EXP_IO_(143) C6
EXP_IO_(144) G10
EXP_IO_(145) G9
EXP_IO_(146) F8
EXP_IO_(147) G8
EXP_IO_(148) B7
EXP_IO_(149) C7
EXP_IO_(150) C5
EXP_IO_(151) D5
EXP_IO_(152) A9
EXP_IO_(153) B9
EXP_IO_(154) A3
EXP_IO_(155) B3
EXP_IO_(156) A4
EXP_IO_(157) B4
EXP_IO_(158) C4
EXP_IO_(159) D4
EXP_IO_(160) E7
EXP_IO_(161) D6
EXP_IO_(162) E6
EXP_IO_(163) E5
EXP_IO_(164) F7
EXP_IO_(165) G7
EXP_IO_(166) C2
EXP_IO_(167) C1
EXP_IO_(168) H8
EXP_IO_(169) H7
EXP_IO_(170) D3
EXP_IO_(171) E4
EXP_IO_(172) G6
EXP_IO_(173) G5

NV28 ○
NV27 ○
NV22 ○
NV21 ○
NV26 ○
NV25 ○
NV24 ○
NV23 ○

IO_L1P_6
IO_L1N_6
IO_L2P_6
IO_L2N_6
IO_L3P_6
IO_L3N_6
IO_L4P_6
IO_L4N_VREF_6
IO_L5P_6
IO_L5N_6
IO_L6P_6
IO_L6N_6
IO_L7P_6
IO_L7N_6
IO_L8P_CC_LC_6
IO_L8N_CC_LC_6
IO_L9P_CC_LC_6
IO_L9N_CC_LC_6
IO_L10P_6
IO_L10N_6
IO_L11P_6
IO_L11N_6
IO_L12P_6
IO_L12N_VREF_6
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IO_L22N_6
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IO_L24N_CC_LC_6
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IO_L29N_6
IO_L30P_6
IO_L30N_6
IO_L31P_6
IO_L31N_6
IO_L32P_6
IO_L32N_6

XC4V-LX60-10FF668C

BANK6
BANK5

BANK6
BANK5

IO_L1P_5
IO_L1N_5
IO_L2P_5
IO_L2N_5
IO_L3P_5
IO_L3N_5
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IO_L5N_5
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IO_L7N_5
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IO_L14N_5
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IO_L15N_5
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IO_L21N_5
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IO_L23N_VRP_5
IO_L24P_CC_LC_5
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IO_L26N_5
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IO_L29N_5
IO_L30P_5
IO_L30N_5
IO_L31P_5
IO_L31N_5
IO_L32P_5
IO_L32N_5

C17
D17 PCI_RESET
C20 PCI_SERR
B20 PCI_PERR
B18 PCI_LOCK
A18 PCI_DEVSEL
D20 PCI_IRDY
D19 PCI_STOP
E17 PCI_TRDY
F17 PCI_FRAME
C21 PCI_IDSEL(0)
B21 PCI_IDSEL(1)
C19 PCI_IDSEL(2)
D18 PCI_IDSEL(3)
A24 PCI_CLK(0)
A23 PCI_CLK(1)
G18 PCI_CLK(2)
G17 PCI_CLK(3)
B24 PCI_PRSNT0_(1)
B23 PCI_PRSNT0_(2)
F18 PCI_PRSNT1_(1)
E18 PCI_PRSNT1_(2)
E21 PCI_PRSNT2_(1)
D21 PCI_PRSNT2_(2)
A20 PCI_PRSNT3_(1)
A19 PCI_PRSNT3_(2)
D22 PCI_CnBE(0)
C22 PCI_CnBE(1)
A22 PCI_CnBE(2)
A21 PCI_CnBE(3)
D24 PCI_PAR
C24 PCI_M66EN
G19 PCI_REQ_SLOT(0)
F19 PCI_REQ_SLOT(1)
E23 PCI_REQ_SLOT(2)
E22 PCI_REQ_SLOT(3)
F20 PCI_GNT_SLOT(0)
E20 PCI_GNT_SLOT(1)
C26 PCI_GNT_SLOT(2)
C25 PCI_GNT_SLOT(3)
D23 PCI_INT_A
C23 PCI_INT_B
H20 PCI_INT_C
G20 PCI_INT_D
G22 PCI_DONE(0)
G21 PCI_DONE(1)
F24 PCI_DONE(2)
F23 PCI_DONE(3)
D26 PCI_BO_(0)
D25 PCI_BO_(1)
H22 PCI_BO_(2)
H21 PCI_BO_(3)
E25 PCI_REQ64_(0)
E24 PCI_REQ64_(1)
G24 PCI_REQ64_(2)
G23 PCI_REQ64_(3)
F26 PCI_ACK64_(0)
E26 PCI_ACK64_(1)
H24 PCI_ACK64_(2)
H23 PCI_ACK64_(3)
G26 TDO(0)
G25 TDO(1)
H26 TDO(2)
H25 TDO(3)

EXP_IO_(63) W2
EXP_IO_(62) W1
EXP_IO_(61) V6
EXP_IO_(60) V5
EXP_IO_(59) W7
EXP_IO_(58) V7
EXP_IO_(57) W4
EXP_IO_(56) W3
EXP_IO_(55) W6
EXP_IO_(54) W5
EXP_IO_(53) Y2
EXP_IO_(52) Y1
EXP_IO_(51) AA4
EXP_IO_(50) AA3
EXP_IO_(49) Y4
EXP_IO_(48) Y3
EXP_IO_(47) Y6
EXP_IO_(46) Y5
EXP_IO_(45) AB1
EXP_IO_(44) AA1
EXP_IO_(43) AC4
EXP_IO_(42) AB4
EXP_IO_(41) AB3
EXP_IO_(40) AB2
EXP_IO_(39) AC5
EXP_IO_(38) AB5
EXP_IO_(37) AC2
EXP_IO_(36) AC1
EXP_IO_(35) AF3
EXP_IO_(34) AE3
EXP_IO_(33) AD2
EXP_IO_(32) AD1
EXP_IO_(31) AF4
EXP_IO_(30) AE4
EXP_IO_(29) AD3
EXP_IO_(28) AC3
EXP_IO_(27) AF6
EXP_IO_(26) AF5
EXP_IO_(25) AA7
EXP_IO_(24) Y7
EXP_IO_(23) AA9
EXP_IO_(22) Y9
EXP_IO_(21) AD5
EXP_IO_(20) AD4
EXP_IO_(19) AE7
EXP_IO_(18) AD7
EXP_IO_(17) AC6
EXP_IO_(16) AB6
EXP_IO_(15) AF8
EXP_IO_(14) AF7
EXP_IO_(13) AA8
EXP_IO_(12) Y8
EXP_IO_(11) Y10
EXP_IO_(10) AA10
EXP_IO_(9) AC7
EXP_IO_(8) AB7
EXP_IO_(7) AC9
EXP_IO_(6) AB9
EXP_IO_(5) AE6
EXP_IO_(4) AD6
EXP_IO_(3) AF9
EXP_IO_(2) AE9
EXP_IO_(1) AD8
EXP_IO_(0) AC8

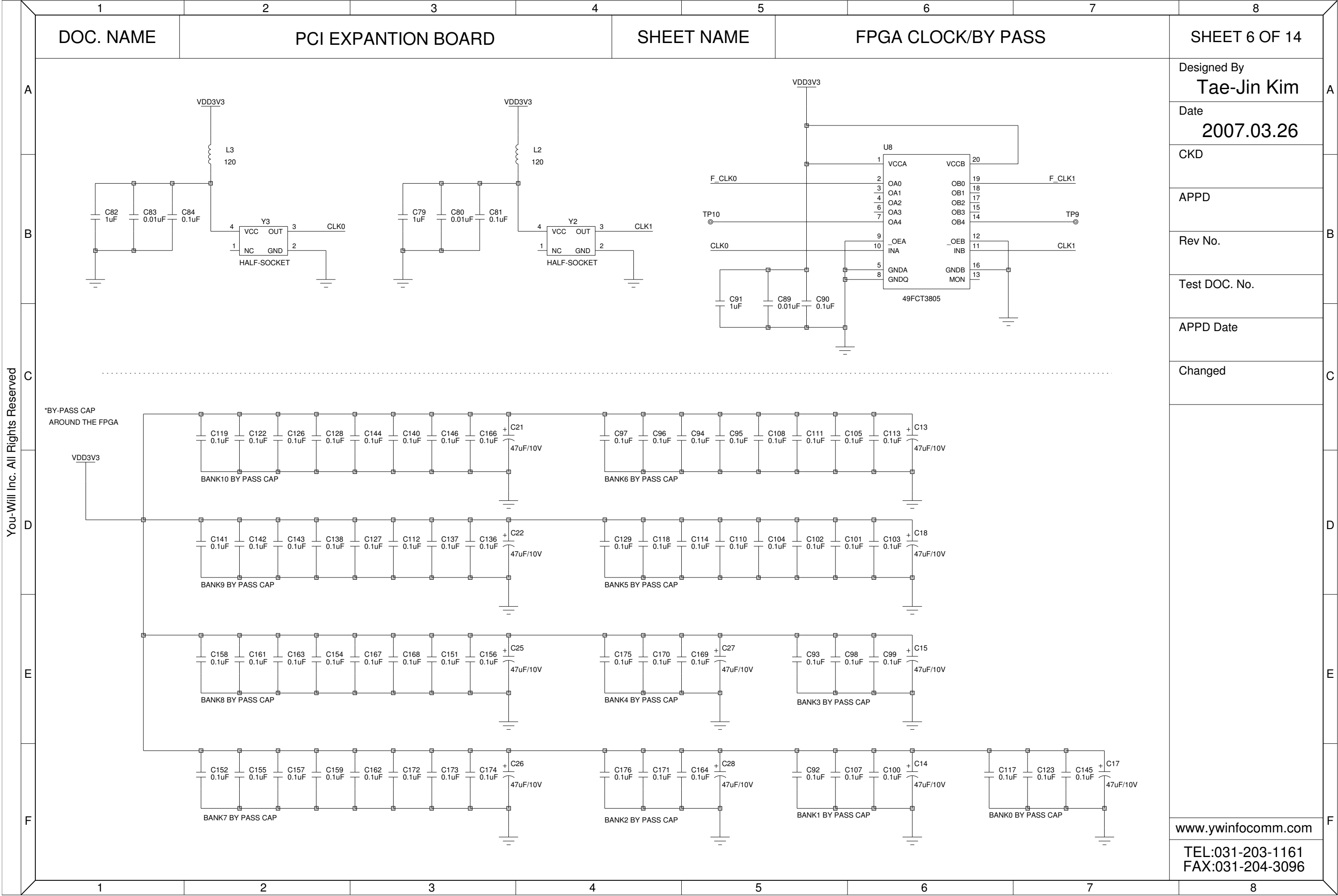
IO_L1P_8
IO_L1N_8
IO_L2P_8
IO_L2N_8
IO_L3P_8
IO_L3N_8
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IO_L4N_VREF_8
IO_L5P_8
IO_L5N_8
IO_L6P_8
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IO_L7P_8
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IO_L32P_8
IO_L32N_8

BANK8
BANK7

BANK8
BANK7

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IO_L1N_7
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IO_L2N_7
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IO_L3N_7
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IO_L30N_SM5_7
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IO_L32N_SM7_7

V21 TCK(0)
V22 TCK(1)
W25 TCK(2)
W26 TCK(3)
W21 TRST(0)
W22 TRST(1)
W23 TRST(2)
W24 TRST(3)
W20 TMS(0)
V20 TMS(1)
Y25 TMS(2)
Y26 TMS(3)
AB24 TDI(0)
AB25 TDI(1)
AA24 TDI(2)
Y24 TDI(3)
AC25
AC26
AB26 ○ NV63
AD25 ○ NV62
AD26 ○ NV68
Y22 ○ NV69
Y23
AC22
AB22
AB23
AA23
AD22
AD23
AC23
AC24
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Y19 PCI_ADDR(2)
W19 RP2 33 PCI_ADDR(3)
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AD21 RP8 33 PCI_ADDR(31)



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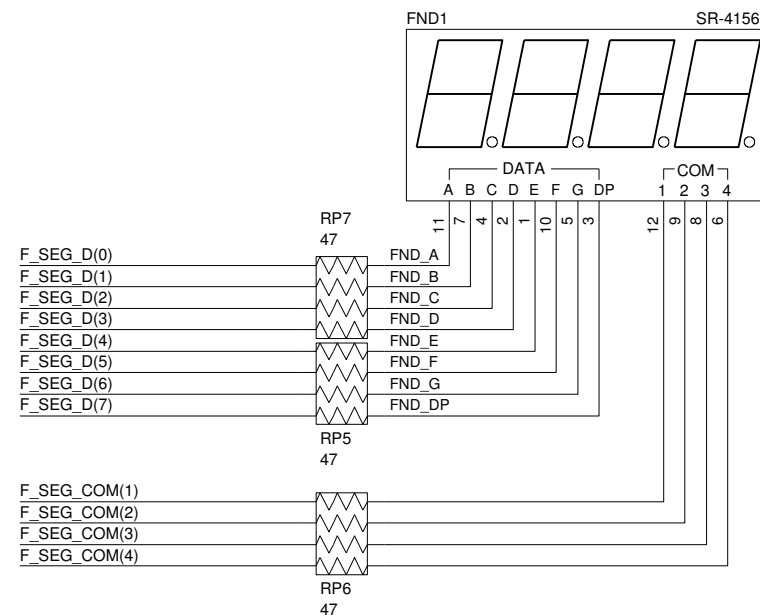
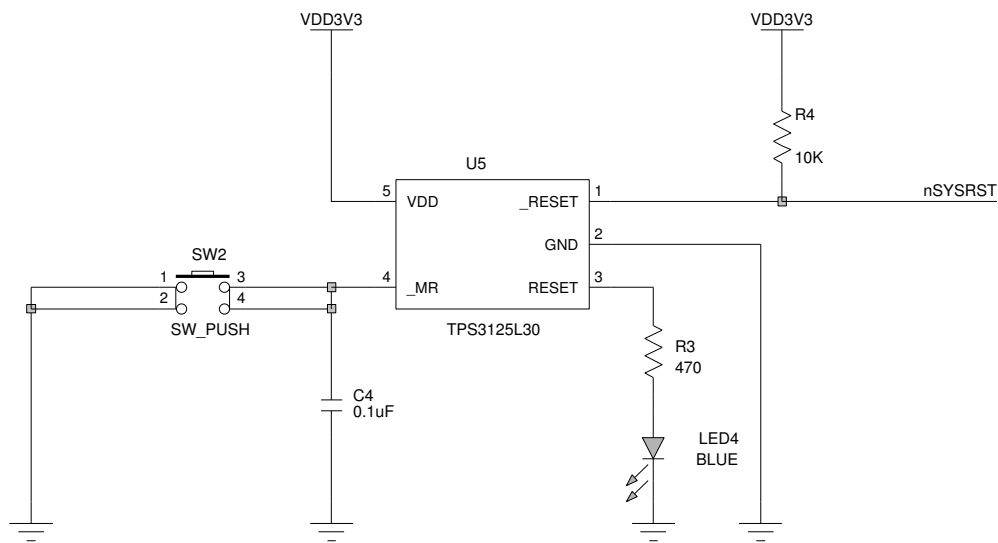
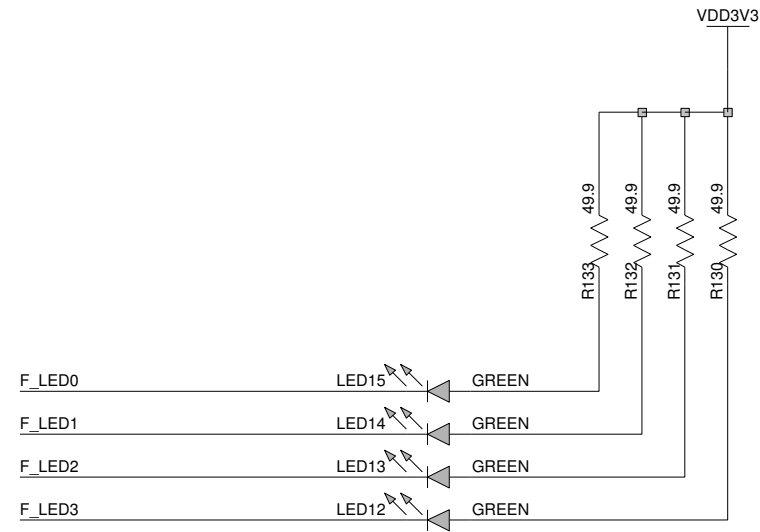
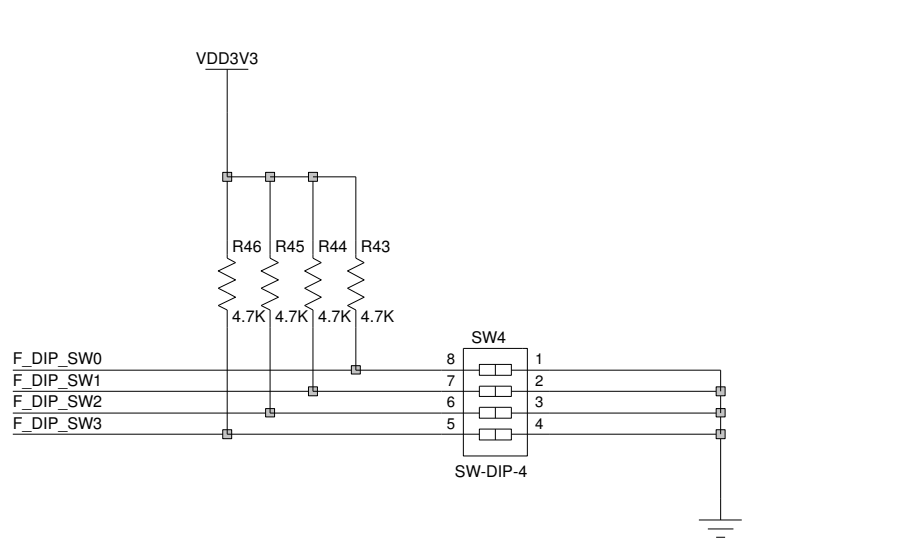
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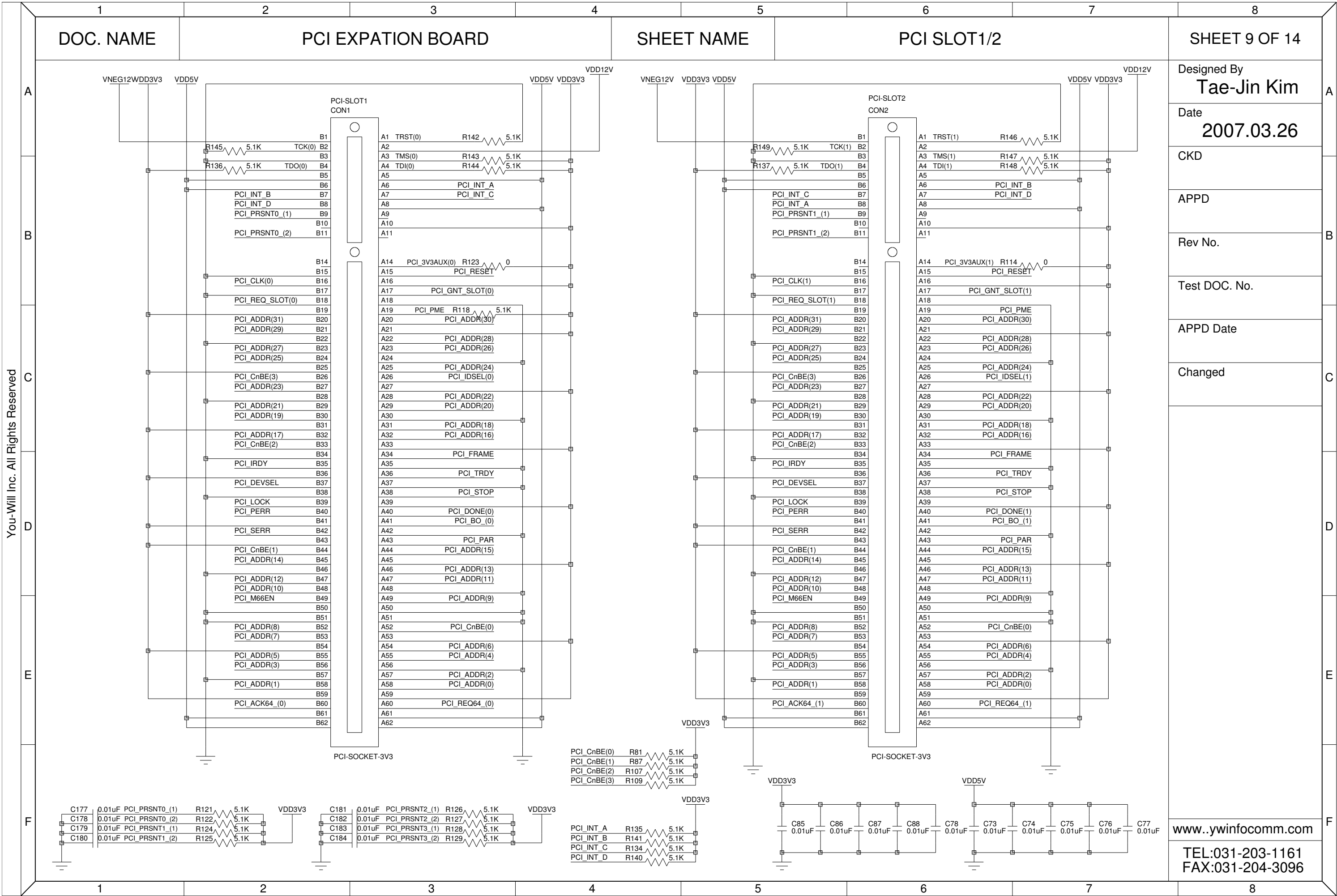
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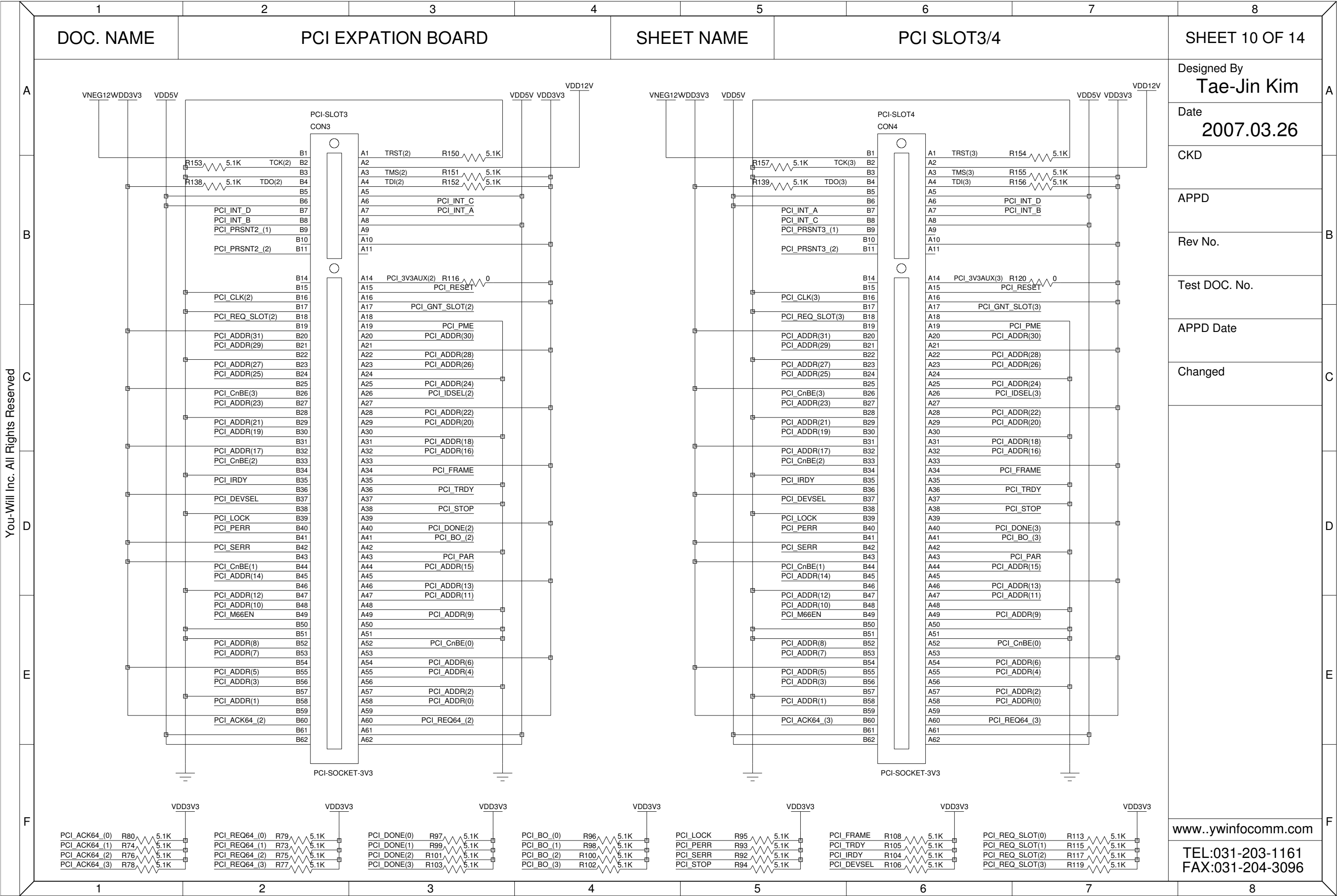
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Designed By Tae-Jin Kim
Date 2007.03.26
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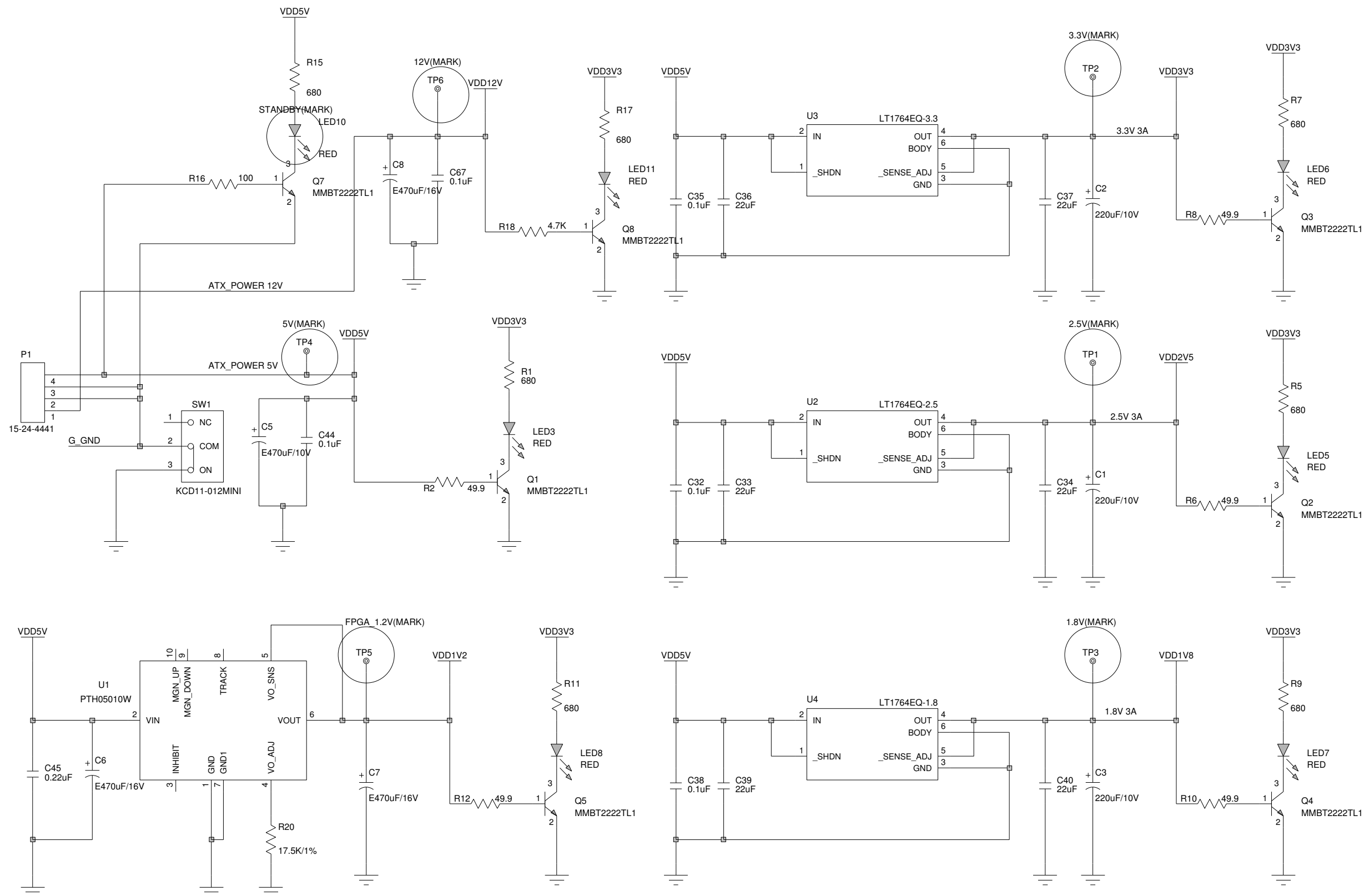






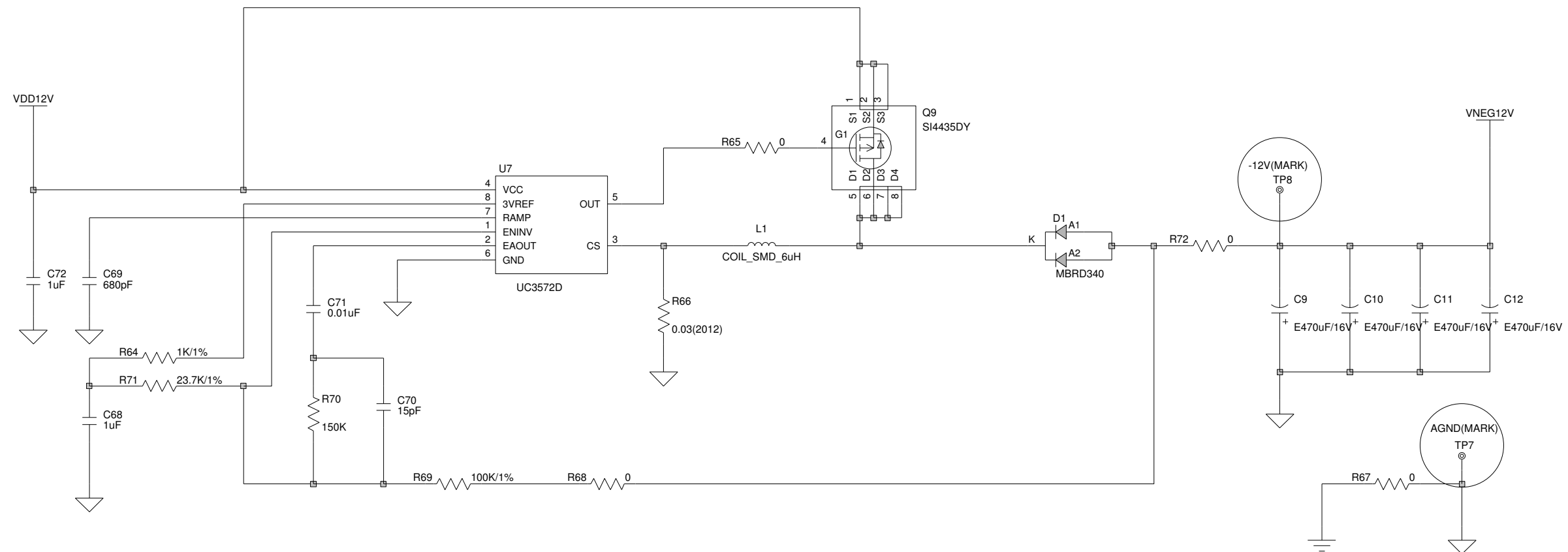
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This figure is a detailed PCB layout for a system ACE, showing various components and their connections. The layout is organized into sections labeled A through F, and columns numbered 1 through 8.

Section A: JTAG VIRTEx-4 FPGA CONFIGURATION

Diagram showing the JTAG configuration circuit. It includes a header (J1) connected to the VDD3V3 supply and the TSTTMS, TSTTCK, TSTTDO, TSTTDI, and G_GND pins of the FPGA (U10).

Section B: PROGRAM RE-DOWNLOAD

Diagram showing the program re-download circuit. It includes a push button (SW6) connected to the VDD3V3 supply and the ACE_NRESET pin of the FPGA (U10).

Section C: XCCACE_TQ144

Diagram showing the XCCACE_TQ144 component, which is a 144-pin connector. It is connected to the VDD3V3 supply and the ACE_NRESET pin of the FPGA (U10).

Section D: HALF-SOCKET

Diagram showing the half-socket circuit. It includes a half-socket (Y1) connected to the VDD3V3 supply and the ACE_NRESET pin of the FPGA (U10).

Section E: SW-DIP-4

Diagram showing the SW-DIP-4 component, which is a 4-pin connector. It is connected to the VDD3V3 supply and the ACE_NRESET pin of the FPGA (U10).

Section F: Power and Ground Connections

Diagram showing the power and ground connections for the system ACE. It includes a VDD3V3 supply connected to the VDD3V3 pin of the FPGA (U10) and the VDD3V3 pin of the system ACE. It also shows ground connections for the system ACE.

System ACE Pin Connections:

- VDD3V3: Connected to the VDD3V3 pin of the system ACE.
- ACE_NRESET: Connected to the ACE_NRESET pin of the system ACE.
- CFD0 to CFD15: Connected to the CFD0 to CFD15 pins of the system ACE.
- CFA0 to CFA10: Connected to the CFA0 to CFA10 pins of the system ACE.
- CFCE0 to CFCE1: Connected to the CFCE0 to CFCE1 pins of the system ACE.
- CFCD0 to CFCD1: Connected to the CFCD0 to CFCD1 pins of the system ACE.
- CFREG, CFWAIT, CFGRSVD: Connected to the CFREG, CFWAIT, CFGRSVD pins of the system ACE.
- CFWE, CFOE: Connected to the CFWE, CFOE pins of the system ACE.
- ERRLED, STATLED: Connected to the ERRLED, STATLED pins of the system ACE.
- NC19 to NC25: Connected to the NC19 to NC25 pins of the system ACE.
- GND1 to GND18: Connected to the GND1 to GND18 pins of the system ACE.
- CFGINT, CFGPROG: Connected to the CFGINT, CFGPROG pins of the system ACE.
- CFGTD0 to CFGTD1: Connected to the CFGTD0 to CFGTD1 pins of the system ACE.
- CFGADDR2 to CFGADDR0: Connected to the CFGADDR2 to CFGADDR0 pins of the system ACE.
- CF_REVSEL(2) to CF_REVSEL(0): Connected to the CF_REVSEL(2) to CF_REVSEL(0) pins of the system ACE.
- CFMODE_ON: Connected to the CFMODE_ON pin of the system ACE.
- TSTTDO to TSTTDI: Connected to the TSTTDO to TSTTDI pins of the system ACE.
- FLASH_TDI, JTAG_TMS, JTAG_TCK, JTAG_TDO, FPGA_TDO: Connected to the FLASH_TDI, JTAG_TMS, JTAG_TCK, JTAG_TDO, FPGA_TDO pins of the system ACE.
- ACE_INT, ACE_PROG: Connected to the ACE_INT, ACE_PROG pins of the system ACE.

System ACE Component Connections:

- MI20A-50PD-SF: Connected to the MI20A-50PD-SF component.
- LED1 (BLUE): Connected to the LED1 (BLUE) pin of the system ACE.
- LED2 (RED): Connected to the LED2 (RED) pin of the system ACE.
- R21, R19: Connected to the R21, R19 resistors.
- C47, C60, C54, C51, C49, C42, C29, C30, C31: Connected to the C47, C60, C54, C51, C49, C42, C29, C30, C31 capacitors.

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		DOC. NAME		PCI EXPANTION BOARD			SHEET NAME		HISTORY			SHEET OF													
A		< HISTORY >										Designed By Tae-Jin Kim		A											
												Date 2007.03.26													
CKD																									
APPD																									
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Test DOC. No.																									
APPD Date																									
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